Multivariate Calculus: Partials, Total Derivatives, and the Chain Rule

Math Foundations — Node 13

1 Definitions

Throughout, $U \subset \mathbb{R}^n$ is open and $f : U \to \mathbb{R}$. We write $\|\cdot\|$ for the Euclidean norm and $e_1, \ldots, e_n$ for the standard basis.

Definition 1 (Partial derivative)**.** Let $x \in U$. The *i-th partial derivative* of f at x is

$$\frac{\partial f}{\partial x_i}(x) := \lim_{h \to 0} \frac{f(x + he_i) - f(x)}{h},$$

provided the limit exists. Equivalently, holding all coordinates fixed except the i-th and viewing f as a univariate function of x_i, the partial is its ordinary derivative.

Definition 2 (Gradient)**.** When all n partials of f at x exist, the *gradient* of f at x is

$$\nabla f(x) := \left(\frac{\partial f}{\partial x_1}(x), \ldots, \frac{\partial f}{\partial x_n}(x) \right) \in \mathbb{R}^n.$$

Definition 3 (Directional derivative)**.** For a unit vector $v \in \mathbb{R}^n$, the *directional derivative* of f at x in direction v is

$$D_v f(x) := \lim_{h \to 0} \frac{f(x + hv) - f(x)}{h}.$$

Definition 4 (Total / Fréchet derivative)**.** f is *differentiable at* $x \in U$ if there exists a linear map $L : \mathbb{R}^n \to \mathbb{R}$ such that

$$\lim_{h \to 0} \frac{\|f(x + h) - f(x) - Lh\|}{\|h\|} = 0.$$

If L exists it is unique; we write $Df(x) := L$. Identifying linear maps $\mathbb{R}^n \to \mathbb{R}$ with row vectors via the inner product, $Lh = \nabla f(x) \cdot h$.

Remark 5. Existence of all partials at x does *not* imply differentiability, nor even continuity. A standard counterexample is $f(x, y) = xy/(x^2 + y^2)$ on the punctured plane, extended by $f(0, 0) = 0$: both partials vanish at the origin, but along $y = x$ the function equals $1/2$, so it is discontinuous there.

2 From partials to the total derivative

Proposition 6 (Continuous partials imply differentiability)**.** *If $\partial f/\partial x_i$ exist in an open neighborhood of x and are continuous at x, then f is differentiable at x with $Df(x)h = \nabla f(x) \cdot h$.*

A proof is in Rudin, Theorem 9.21. In ML we work almost exclusively with C^1 functions, so this proposition lets us identify differentiability with the gradient picture without further worry.

3 The chain rule

The central result of this node is the multivariate chain rule along a curve.

Theorem 7 (Multivariate chain rule, scalar codomain)**.** *Let $U \subset \mathbb{R}^n$ be open, $f : U \to \mathbb{R}$ differentiable at $a \in U$, and $\gamma : I \to U$ a curve on an open interval $I \subset \mathbb{R}$ with $\gamma(t_0) = a$ and γ differentiable at t_0. Then $f \circ \gamma : I \to \mathbb{R}$ is differentiable at t_0 and*

$$(f \circ \gamma)'(t_0) \;=\; \nabla f(a) \cdot \gamma'(t_0) \;=\; \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(a)\, \gamma_i'(t_0).$$

Proof. Set $L := Df(a)$, identified with $\nabla f(a) \in \mathbb{R}^n$ via $Lw = \nabla f(a) \cdot w$. Differentiability of f at a means: for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$\|w\| < \delta \implies \bigl|f(a + w) - f(a) - Lw\bigr| \le \varepsilon \|w\| . \tag{1}$$

Equivalently, there is a function $\varphi : \mathbb{R}^n \to \mathbb{R}$ with $\varphi(w) \to 0$ as $w \to 0$ and

$$f(a + w) - f(a) = Lw + \varphi(w) \|w\| , \qquad \varphi(0) := 0. \tag{2}$$

Differentiability of γ at t_0 similarly gives: there is a function $\psi : I - t_0 \to \mathbb{R}^n$ with $\psi(s) \to 0$ as $s \to 0$ and

$$\gamma(t_0 + s) - \gamma(t_0) = s\, \gamma'(t_0) + s\, \psi(s), \qquad \psi(0) := 0. \tag{3}$$

Set $w(s) := \gamma(t_0 + s) - a$. By (3), $w(s) = s(\gamma'(t_0) + \psi(s))$, so

$$\|w(s)\| \le |s|\bigl(\bigl\|\gamma'(t_0)\bigr\| + \|\psi(s)\|\bigr). \tag{4}$$

Pick s small enough that $\gamma(t_0 + s) \in U$ and $\|w(s)\| < \delta$ (possible since γ is continuous at t_0). Substituting $w = w(s)$ into (2):

$$f(\gamma(t_0 + s)) - f(a) = Lw(s) + \varphi(w(s)) \|w(s)\| .$$

By linearity of L and (3),

$$Lw(s) = L\bigl(s\gamma'(t_0) + s\psi(s)\bigr) = s\, L\gamma'(t_0) + s\, L\psi(s).$$

Therefore

$$\frac{f(\gamma(t_0 + s)) - f(\gamma(t_0))}{s} = L\gamma'(t_0) + L\psi(s) + \varphi(w(s)) \cdot \frac{\|w(s)\|}{s}. \tag{5}$$

We send $s \to 0$ term by term. The first term $L\gamma'(t_0) = \nabla f(a) \cdot \gamma'(t_0)$ is the target. For the second, L is continuous (every linear map on $\mathbb{R}^n$ is), and $\psi(s) \to 0$, so $L\psi(s) \to 0$. For the third, $w(s) \to 0$ by (4), so $\varphi(w(s)) \to 0$, while $|\,\|w(s)\| /s|$ is bounded above by $\|\gamma'(t_0)\| + \|\psi(s)\|$, which stays bounded near $s = 0$; hence the product tends to 0.

Combining, the limit of the left-hand side of (5) exists and equals $\nabla f(a) \cdot \gamma'(t_0)$. That limit is by definition $(f \circ \gamma)'(t_0)$, completing the proof. $\square$

Remark 8 (The directional-derivative corollary). Take $\gamma(t) = x + tv$ with $v \in \mathbb{R}^n$ a unit vector. Then $\gamma'(0) = v$, and Theorem 7 specializes to

$$D_v f(x) = (f \circ \gamma)'(0) = \nabla f(x) \cdot v.$$

This identity is the basis of the Cauchy–Schwarz argument that the gradient points in the direction of steepest ascent: $D_v f(x) \le \|\nabla f(x)\|$ with equality at $v = \nabla f(x)/ \|\nabla f(x)\|$.

4 Worked example

Let $f(x, y) = x^2y + \sin(xy)$. Computing partials by holding the other variable fixed:

$$\frac{\partial f}{\partial x} = 2xy + y\cos(xy), \qquad \frac{\partial f}{\partial y} = x^2 + x\cos(xy).$$

At $(1, 2)$: $\partial_x f = 4{+}2\cos 2$, $\partial_y f = 1{+}\cos 2$. So $\nabla f(1, 2) = (4{+}2\cos 2,\ 1{+}\cos 2)$. For $v = (1, 1)/\sqrt{2}$,

$$D_v f(1, 2) = \frac{(4 + 2\cos 2) + (1 + \cos 2)}{\sqrt{2}} = \frac{5 + 3\cos 2}{\sqrt{2}}.$$

The companion script `code.py` confirms these values numerically.